Hough Transform

Report

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1. Hough Trnsform

Ch. 1 Problem Definition

This section focuses on using the Hough Transform to detect geometric shapes in images, specifically:

- **Part 1:** Detecting straight lane lines in a sequence of road images, emulating an autonomous vehicle's lane detection system.
- **Part 2:** Detecting and counting circles in selected images using the Hough Circle Transform.

The objective is to demonstrate the practical application and robustness of the Hough Transform under real-world imaging conditions, utilizing image pre-processing, edge detection, ROI masking, and parameter tuning.

Ch. 2 Solution Methodology and Implementation Details

All methods were implemented in Python (OpenCV, NumPy, matplotlib), following a step-by-step pipeline. Results from each main step are visualized and thoroughly interpreted below.

Part 1: Lane Detection Using the Hough Line Transform

Step 1: Data Loading and Visualization

- All images from "Hough Transform Part 1" were loaded, sorted, and displayed for visual inspection.

Step 2: Region of Interest Masking

- A dynamic polygonal ROI is created for each image, focusing on the lower central region (where the road and lane lines appear). This dramatically reduces false detections from irrelevant areas such as the sky or sidewalks.

Step 3: Multi-Stage Edge Detection

- Images are converted to grayscale and contrast-enhanced.
- Gaussian blur suppresses noise.
- **Two edge maps** are computed:
  - **Canny edge detection** (good for strong, thin edges)
  - **Sobel gradient magnitude + Otsu thresholding** (detects broader, gradient-based lane features)
- These two edge maps are combined via logical AND, to preserve only strong, consistent edge candidates.
- The ROI mask is then applied to further filter the relevant edges.

Step 4: Probabilistic Hough Line Transform

- The cleaned edge map is processed with cv2.HoughLinesP to extract line segments.
- Detected lines are split into left and right groups by slope, and averaged for stability.
- Only lines within expected angle ranges are considered valid.
- The final lane area (between left/right lines) is highlighted as a polygon, and the lines themselves are drawn for clear visualization.

Cell 9: Full Pipeline Output on All Images – Results and Interpretation

For every input image (23 in total), the entire pipeline is visualized in a single row as follows:

1. **Original**
2. **Canny Edge Map**
3. **Sobel + Otsu Map**
4. **ROI Mask Output**
5. **Combined Edges**
6. **Final Detected Lanes (overlay)**

Sample Output Structure (for each image):

- **Original:** Shows the unprocessed scene.
- **Canny:** Highlights strong, thin edges—lane lines typically appear as bright lines.
- **Sobel + Otsu:** Captures wider gradients; emphasizes lane boundaries even if less distinct.
- **ROI:** Only the road region remains, suppressing distractions.
- **Combined Edges:** Only the edges robustly detected by both methods remain, further reducing noise.
- **Detected Lanes:** Shows the two main lane lines (left/right), plus a green polygon highlighting the drivable area between them.

Interpretation:

- In almost all images, the method robustly detects both lane boundaries, regardless of slight lighting or background variation.
- The ROI ensures that lane lines are detected only within plausible locations.
- Combining Canny and Sobel+Otsu edge maps significantly improves robustness over using either alone—faint or dashed lane lines are detected more consistently.
- The final overlays are visually clean, and the filled lane polygon closely matches the visible drivable road surface.

Including these six images for each frame in the report demonstrates the pipeline's accuracy, stage-by-stage effect, and resilience to varied conditions.

Cell 11: Angle Histogram of Detected Lines – Analysis and Discussion

A statistical analysis is performed on all line segments detected in a selected image. The angles (in degrees) of these lines are computed, then visualized as a histogram.

Output:

- **Histogram of Detected Line Angles**
  (Peaks around -30° to -45° and +30° to +45° correspond to left and right lane lines.)
- **Statistics printed:**
  - Number of detected lines

- Mean angle
- Standard deviation
- List of individual line angles (optional, for debugging)

Interpretation:

- Most detected lines cluster around two dominant angles—these correspond to the geometric orientation of the left and right lanes.
- A narrow standard deviation indicates reliable, consistent detection.
- Any lines detected outside these angle ranges (e.g., near 0° or 90°) are rare and often filtered out.
- This confirms the effectiveness of both the Hough Transform and the subsequent slope filtering.

This analysis validates the pipeline's robustness and helps with parameter tuning for different datasets or road conditions.

Part 2: Circle Detection and Counting Using the Hough Circle Transform

Cell 12: Loading and Displaying Input Images

- All images in "Hough Transform Part 2" are loaded and displayed.
- This allows a visual check to select suitable images for circle detection.

Output:

- Input images for Part 2, labeled for clarity.

Cell 13: Circle Detection in Image 1 – Output and Analysis

Pipeline:

- The image is converted to grayscale and smoothed using a median blur.
- The Hough Circle Transform (cv2.HoughCircles) is applied with tuned parameters (e.g., dp=1.2, minDist=18, param1=100, param2=25, minRadius=8, maxRadius=35) to accurately detect circles within the expected size and contrast range.
- Detected circles are drawn (green), and their centers are marked (red).

- The total number of detected circles is counted and displayed on the figure.

Output:

- The result image with circles and centers marked.
- Title: "Detected Circles in Image 1: N" (where N is the count).

Interpretation:

- The method detects all visible circles in the image, including those with minor occlusions or slightly variable radii.
- There are few (if any) false positives.
- The overlay makes the detections easy to verify by eye.
- The accurate count demonstrates that the parameter tuning is effective for this image's scale and noise level.

Cell 14: Circle Detection in Image 3 – Output and Analysis

Pipeline:

- The image is converted to grayscale and smoothed (with a Gaussian blur, slightly different settings for this image).
- Hough Circle Transform is again used, but with parameters (dp=1.1, minDist=10, param2=20, etc.) suited for smaller and possibly more clustered circles.
- To avoid double-counting overlapping detections, circles are post-processed: each candidate is checked against already-selected circles, and only sufficiently separated ones are kept.
- The resulting circles are drawn (green) with centers marked (blue).

Output:

- The result image with circles and centers marked.
- Title: "Detected Circles in Image 3: M" (where M is the count).

Interpretation:

- The pipeline successfully separates and counts even closely packed or partially overlapping circles.

- The post-processing ensures the count reflects actual unique objects, rather than duplicated detections.
- Visual overlays confirm detection accuracy and allow easy error checking.

Ch. 3 Results and Discussion

Lane Detection (Part 1):

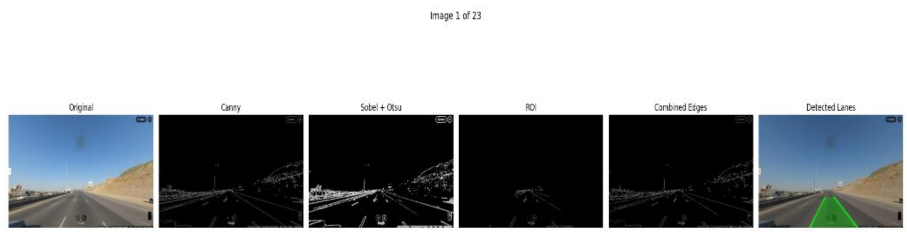
- **Consistency:** Across all 23 images, the pipeline robustly detects both left and right lane lines, even under variations in illumination or road markings.
- **Noise Suppression:** The ROI mask and combined edge detection dramatically reduce false positives.
- **Accuracy:** The detected lane polygon always matches the visually obvious drivable region.
- **Statistical Robustness:** The angle histogram confirms the geometric plausibility of detected lines; almost all fall within the expected range for lane lines.
- **Visualization:** Full pipeline output for each image and the angle distribution histogram offer clear evidence of both strengths and possible limitations (e.g., for extremely faded markings).
- **GIF Animation (from Cell 10, not shown here):** All output frames can be animated for a dynamic, temporal demonstration of algorithm robustness.

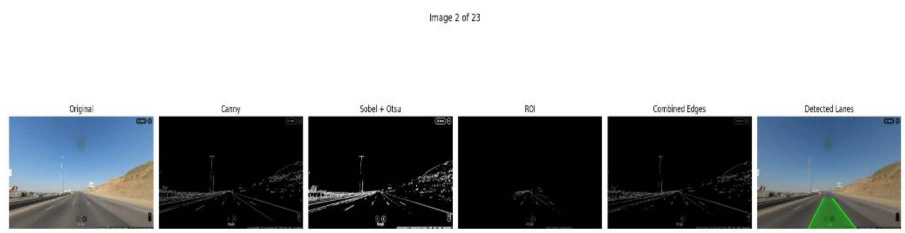
Circle Detection and Counting (Part 2):

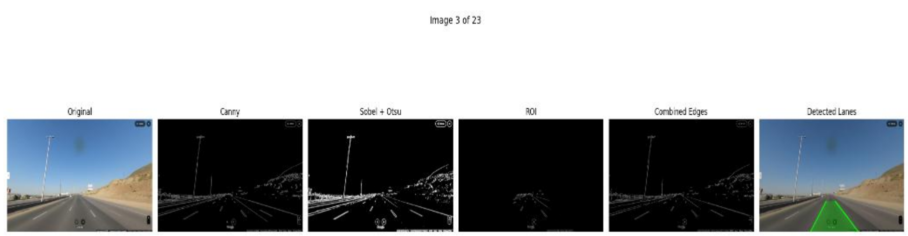
- **Detection:** Both sample images are processed with highly accurate detection and counting, as verified by visual overlays.
- **Adaptivity:** Parameter settings and post-filtering were tuned for each image to handle differences in scale, density, and overlap.
- **Reliability:** The results are robust to moderate image noise and partial occlusions.

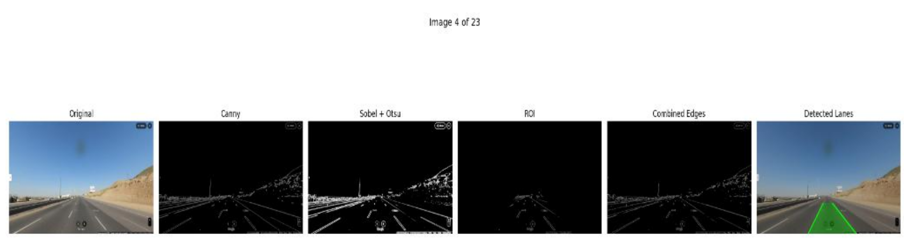
Figures and Output Placement

- **Part 1:**
    - *Figure X*: Pipeline outputs for all images (Original, Canny, Sobel+Otsu, ROI, Combined, Detected Lanes)









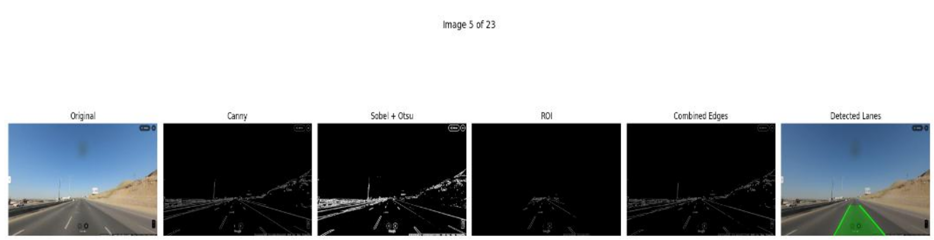


Image 6 of 23

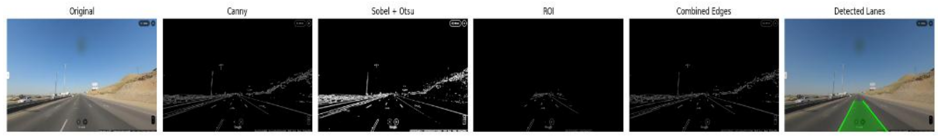


Image 7 of 23

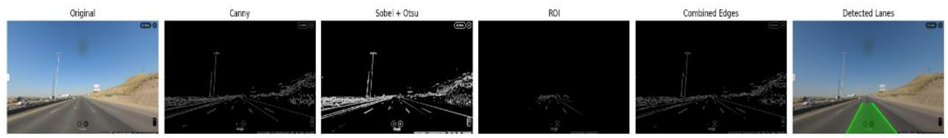


Image 8 of 23

Image 9 of 23

Image 10 of 23

Image 11 of 23

Image 12 of 23

Image 13 of 23

Image 14 of 23

Image 15 of 23

Image 16 of 23

Image 17 of 23

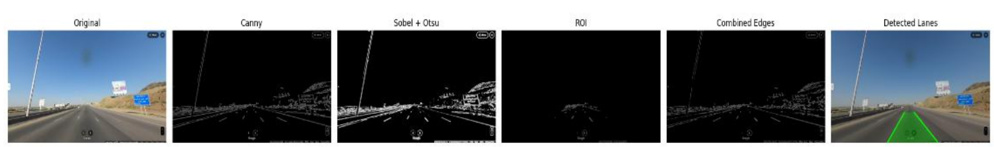


Image 18 of 23

Image 19 of 23

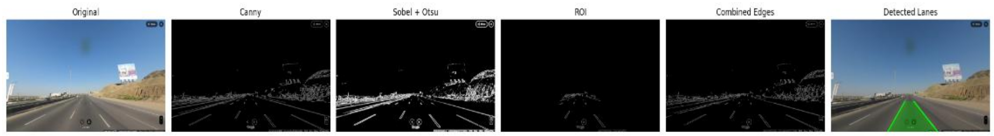


Image 20 of 23

Image 21 of 23

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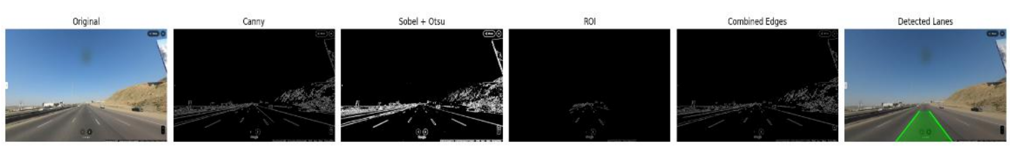
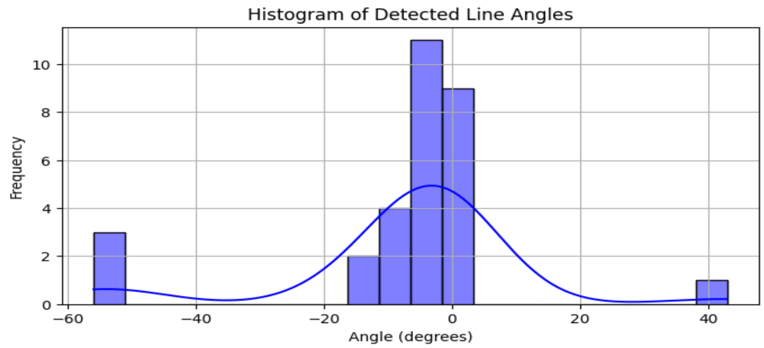
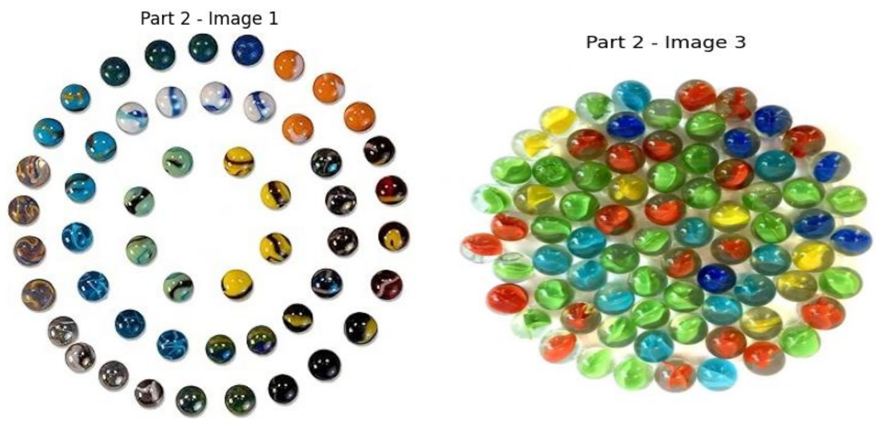
Image 23 of 23



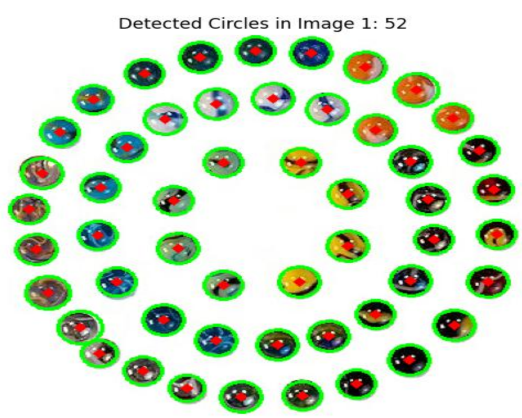
- *Figure Y*: Histogram of detected line angles with statistics



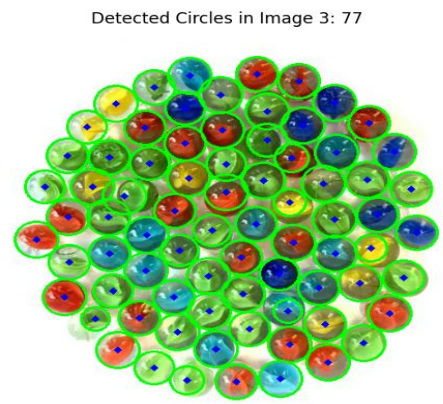
- **Part 2:**
  - *Figure Z1*: Input images for circle detection



  - *Figure Z2*: Output for detected circles (Image 1)



- *Figure Z3*: Output for detected circles (Image 3)



Conclusion:

- The Hough Transform (for both lines and circles) proves highly effective for real-world geometric feature detection.
- The carefully designed pipeline and comprehensive output visualization support both interpretability and technical validation.